



Artificial Intelligence Foundations (CIS4049-N)

Week 1 – What's AI – Part 4

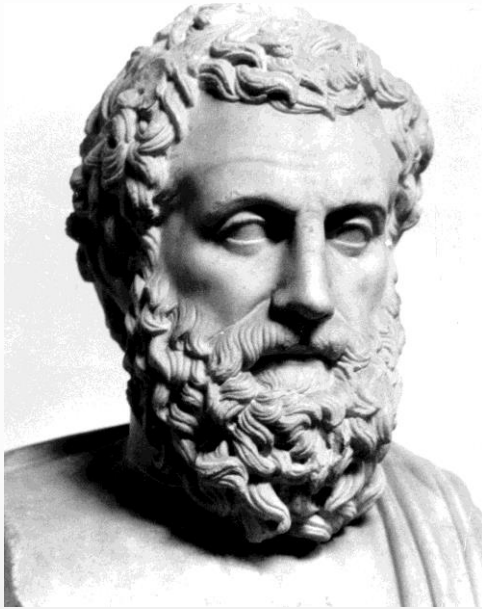
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Two AI Traditions



- AI has always swung back and forth between these two intellectual traditions
- Deep philosophical differences, but deeper connections (McCulloch/Pitts, AlphaGo)?

A melting pot

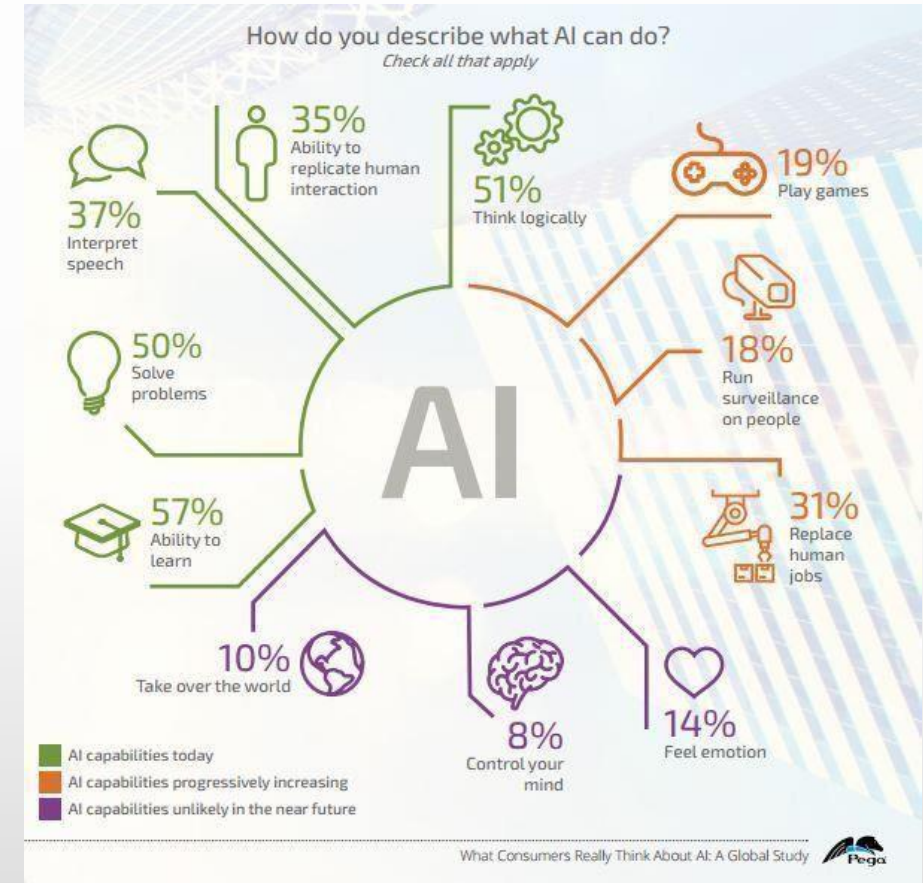
- Bayes rule (Bayes)
- Least squares regression
- First-order logic
- Maximum likelihood
- Artificial neural networks
- Minimax games
- Stochastic gradient descent
- Uniform cost search
- Value iteration



What can AI do?

What Can AI Do?

- Play a decent game of Jeopardy? ✓
- Win against any human at chess? ✓
- Win against the best humans at Go? ✓
- Play a decent game of tennis? ✓
- Grab a particular cup and put it on a shelf? ✓
- Unload any dishwasher in any home? ✗
- Drive safely along the highway? ✓
- Drive safely along Oxford Street in London? ✗
- Buy a week's worth of groceries on the web? ✓
- Buy a week's worth of groceries in a crowded market? ✗
- Discover and prove a new mathematical theorem? ?
- Perform a surgical operation? ?
- Translate spoken Chinese into spoken English in real time? ✓
- Write an intentionally funny story? ✗



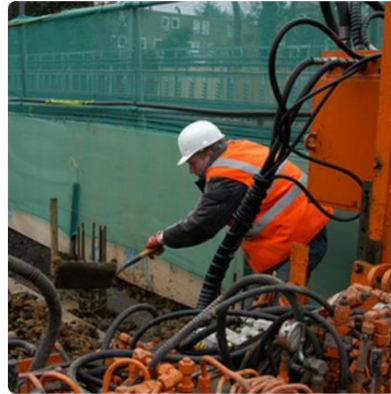
Natural Language

- **Speech technologies** (e.g. Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
 - Dialog systems
- **Language processing technologies**
 - Question answering
 - Machine translation
 - Web search
 - Text classification, spam filtering, etc.

Computer Vision



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



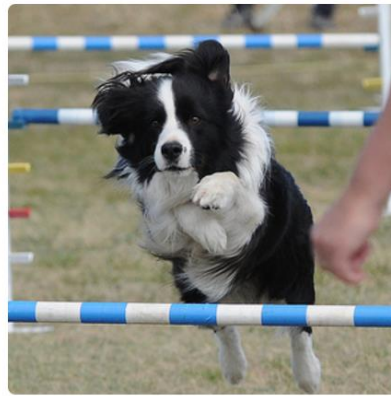
"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."

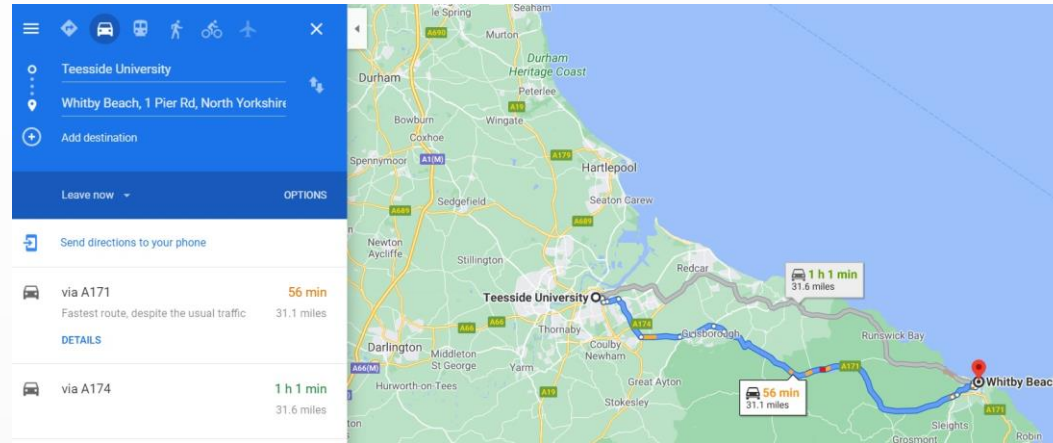


"young girl in pink shirt is swinging on swing."



"man in blue wetsuit is surfing on wave."

Tools for Predictions & Decisions



Game Agents

- Classic Moment: May, '97: Deep Blue vs. Kasparov
 - First match won against world champion
 - “Intelligent creative” play
 - 200 million board positions per second
 - Humans understood 99.9 of Deep Blue's moves
 - Can do about the same now with a PC cluster



Credit: Text from Bart Selman, image from IBM's Deep Blue pages

- **1996: Kasparov Beats Deep Blue**

“I could feel --- I could smell --- a new kind of intelligence across the table.”

- **1997: Deep Blue Beats Kasparov**

“Deep Blue hasn't proven anything.”

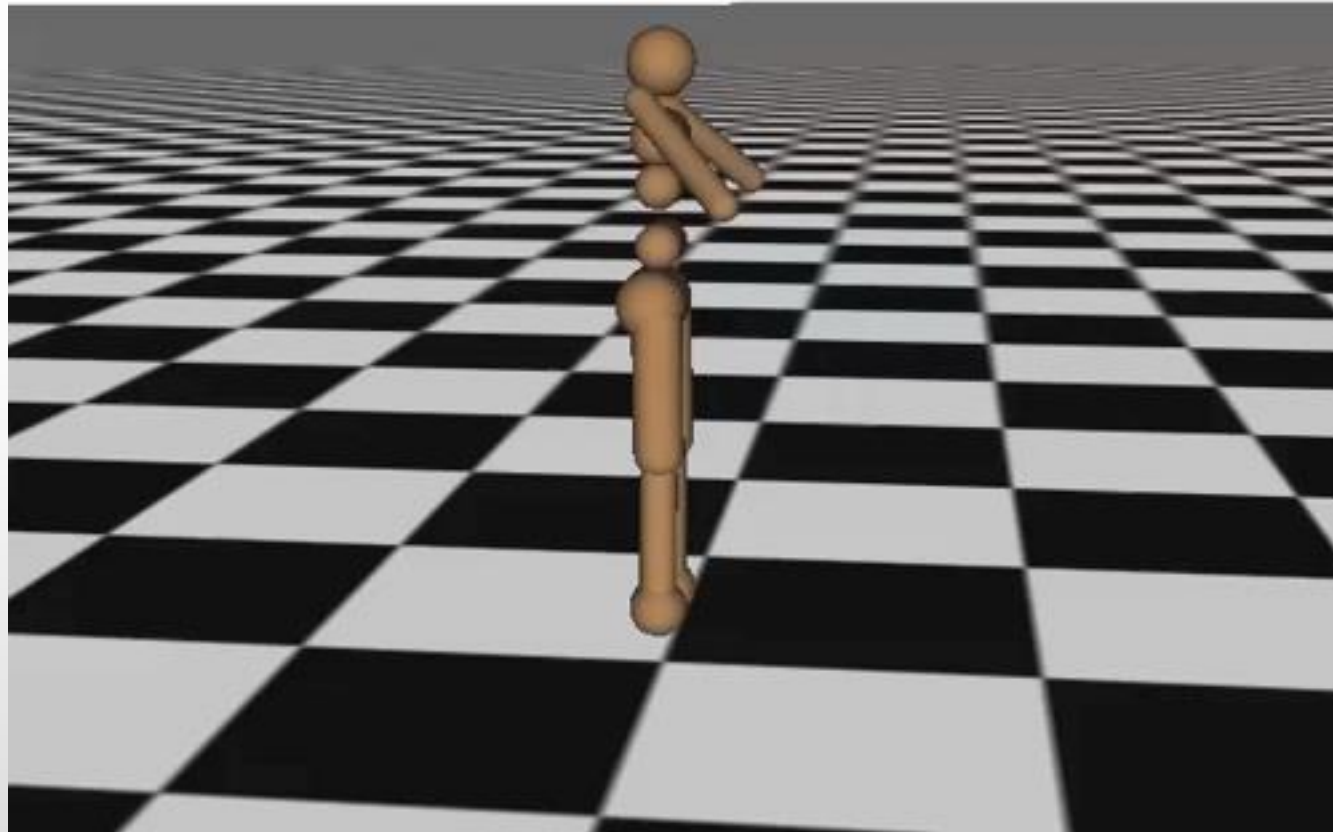
Game Agents

- Reinforcement



Simulated Agents

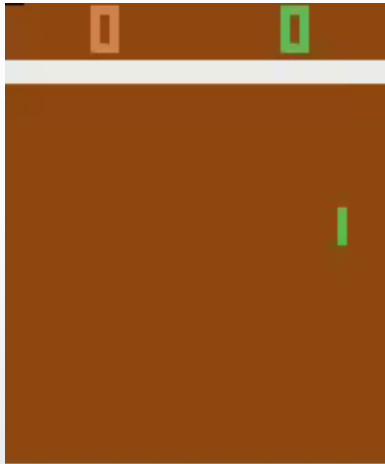
Iteration 0



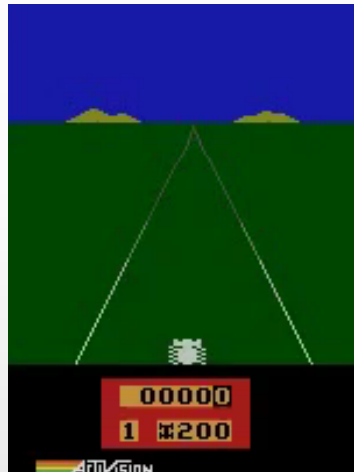
[Schulman, Moritz, Levine, Jordan, Abbeel, ICLR 2016]

Game Agents

- Reinforcement learning



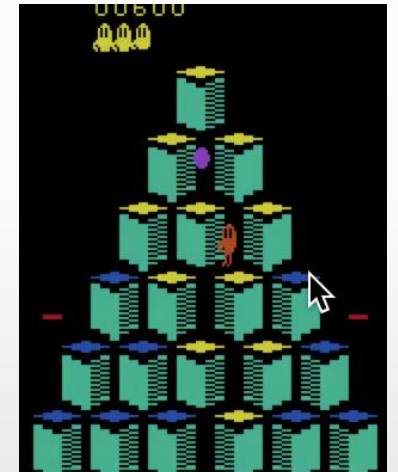
Pong



Enduro



Beamrider



Q*bert

Robotics

- **Robotics**

- Part mechanical engineering
- Part AI
- Reality much harder than simulations!

- **Technologies**

- Vehicles
- Rescue
- Help in the home
- Lots of automation



Credit: Images from UC Berkeley, Boston Dynamics, RoboCup, Google

Human-AI Interaction – Self-Driving

